DOOB'S PROOF OF MLE CONSISTENCY

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Abstract. A short presentation of Doob's proof of the consistency of Fisher's maximum likelihood estimator.

1. Introduction.

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered more than a century ago. This method is typically subsumed under the rubric of M-estimators, and derived via standard techniques dating back to Cramér [6] and Wald [24]. For a more recent treatment, van der Vaart [23] is standard. Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found Doob's consistency proof [9] of MLE convergence more motivated and accessible to students, especially to students already exposed to information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The aim of this note is to present this proof, together with some historical terminology remarks, in a self-contained manner. By using the law of large numbers in its strong form, we derive MLE consistency in its strong form.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2. Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X. We suspect the density of X is $p(X, \theta^*)$, one of a parametric family $p(X, \theta)$ of densities, and we wish to recover θ^* from $X_1, X_2, \dots$.

Stated in this manner, all we have to work with are two items: the true expectation E_* and the family $p(X, \theta)$. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over or knowledge of E_*. Two issues immediately spring to mind. Since we haven't defined "density", how is θ^* defined? Once defined, how do we use $X_1, X_2, \dots$ to recover θ^*?

To this end, we define the *surprisal* [18, 21]

$$S(X, \theta) = \log(1/p(X, \theta)). \tag{1}$$

This terminology is intuitive: low probability equals high surprisal. Then it is natural to characterize θ^* as a minimizer of the expected surprisal $E_*(S(X, \theta))$. More precisely, we say *the parametric family has a true parameter* if $S(X, \theta)$ is integrable for all θ, $E_*(S(X, \theta))$ has a minimizer θ, and the density $p(X, \theta)$ of minimizers is uniquely determined. When the parametric family has a true parameter, any minimizer is a *true parameter*.

Equivalently, given θ^*, the *relative information* is

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)). \tag{2}$$

Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ and $I(\theta^*, \theta) = 0$ implies $p(X, \theta) = p(X, \theta^*)$. The relative information I appears in many fields, in different forms. Because of widely differing terminology, below we give an account of its history.

A special case is when there is[1] a background expectation E such that

$$E_\theta(f(X)) = E(p(X,\theta)f(X))$$

is an expectation for all θ. When this holds and $E_{\theta^*} = E_*$, the relative information has the familiar form

$$I(\theta^*,\theta) = E\left(p(X,\theta^*)\log\left(\frac{p(X,\theta^*)}{p(X,\theta)}\right)\right),$$

and θ^* is[2] a true parameter [7, 8].

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X,\theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X,\theta) = \log(1/\theta_X)$, and

$$I(\theta^*,\theta) = \sum_{i=1}^d \theta_i^* \log(\theta_i^*/\theta_i). \tag{3}$$

Turning back to the general setting, the *likelihood* and *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^n p(X_k,\theta), \qquad \bar{S}_n(\theta) = \frac{1}{n}\log(1/L_n(\theta)) = \frac{1}{n}\sum_{k=1}^n S(X_k,\theta).$$

The mean surprisal is also called the negative-log-likelihood.

By definition, θ^* minimizes $E_*(S(X,\theta))$. By the law of large numbers, $\bar{S}_n(\theta)$ converges to $E_*(S(X,\theta))$ as $n \to \infty$, with probability one. Hence it is natural to estimate θ^* by a minimizer of $\bar{S}_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \ge 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the outcome, maximizing $L_n(\theta)$ over all θ. Under **B** below, with probability one, an MLE sequence $\hat{\theta}_n$ exists for sufficiently large n.

Both $\hat{\theta}_n$ and θ^* depend on the choice of $p(X,\theta)$: With a different parametric family, the MLE sequence and the true parameter will be different.

3. Consistency

To achieve consistency, there are three natural assumptions, the first of which is

A. *(Identifiability):* $\theta \neq \theta'$ implies $p(X,\theta) \neq p(X,\theta')$, hence $I(\theta^*,\theta) > 0$. Without identifiability, one cannot hope to extract θ from $p(X,\theta)$. With identifiability, the true parameter is unique.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \le c$ are compact subsets of the parameter space, for every level c. Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the second assumption

B. *(Properness):* There is an $N \ge 1$, depending on the outcome, and a proper $F(\theta)$, not depending on the outcome, such that, with probability one,

$$F(\theta) \le \bar{S}_n(\theta), \qquad \text{for all } \theta \text{ and } n \ge N. \tag{4}$$

We will need the convergence of $\bar{S}_n(\theta)$ to $E_*(S(X,\theta))$ across θ, in the sense

$$\bar{S}_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{5}$$

with probability one. For this we need the third assumption

[1] While $E_\theta(1) = 1$, E need not satisfy $E(1) = 1$, $E(1) = \infty$ is allowed.

[2] Apply Jensen's inequality with the strictly convex $\log(1/p)$.

C. *(Continuity):* The parameter space is separable and there is a continuous $f(\theta,\theta')$ and an integrable $g(X)$ satisfying

$$|S(X,\theta) - S(X,\theta')| \le f(\theta,\theta')g(X) \qquad \text{and} \qquad f(\theta,\theta) = 0.$$

Then a separability argument [19] establishes (5) with probability one.

Under **B**, **C**, with probability one, an MLE $\hat\theta_n$ exists[3] for $n \ge N$. **B**, **C** are typically checkable for *exponential families* [5, 15, 19, 23], the densities $p(X,\theta)$ where $S(X,\theta)$ is a sum of products of the form $f(\theta)g(X)$.

Theorem (Consistency)**.** *Let X_1, X_2, … be independent and identically distributed, and let $p(X,\theta)$ be a parametric family with a true parameter θ^*. Under **A**, **B**, **C**, with probability one, an MLE sequence $\hat\theta_n$ may be defined for all but finitely many n, and any such sequence converges to θ^* as $n \to \infty$.*

Proof. By the law of large numbers,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{6}$$

with probability one. Fix an outcome for which (4), (5) and (6) hold. We show $\hat\theta_n \to \theta^*$ as $n \to \infty$ for this outcome.

By (4),

$$F(\hat\theta_n) \le \bar S_n(\hat\theta_n) \le \bar S_n(\theta^*), \qquad \text{for } n \ge N.$$

Since the right side is bounded as $n \to \infty$, and $F(\theta)$ is proper, the sequence $\hat\theta_n$ lies in a compact set in the parameter space.

Now suppose $\hat\theta_n \to \theta$ along a subsequence. By **C**,

$$\bar S_n(\theta) - \bar S_n(\theta^*) \le \bar S_n(\theta) - \bar S_n(\hat\theta_n) \le f(\theta,\hat\theta_n)\cdot\frac{1}{n}\sum_{k=1}^{n} g(X_k). \tag{7}$$

Since the left side of (6) converges to $E_*(g(X))$, and $f(\theta,\hat\theta_n) \to 0$ along a subsequence, the right side of (7) goes to zero along a subsequence. By (2) and (5), the left side of (7) converges to $I(\theta^*,\theta)$, hence $I(\theta^*,\theta) \le 0$. But $I(\theta^*,\theta) \ge 0$, hence $I(\theta^*,\theta) = 0$. From **A**, this implies $\theta = \theta^*$. □

4. Terminology History

The relative information $I(\theta^*,\theta)$ is often called "relative entropy" [5, 15]. The term "entropy", originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann's 1877 combinatorial interpretation [3] and Planck's 1901 paper [17], flipped Boltzmann's sign convention, writing the entropy H as done today.

Shannon, in his 1948 paper [20], independently defined

$$H(p) = -\sum_{i=1}^{d} p_i \log p_i.$$

Von Neumann had earlier defined quantum entropy in his 1932 book [16], and when Shannon consulted him around 1939–1940 about terminology for H, von Neumann suggested "entropy", which Shannon adopted [22]. However, in his remark "H is then, for example, the H in Boltzmann's H theorem", Shannon conflates his H with Boltzmann's H.

The relative information $I(\theta^*,\theta)$ is jointly convex in the densities $p(X,\theta^*)$ and $p(X,\theta)$, a fact that is apparent in the special case (3), whereas $H(p)$ above is strictly concave in p. Because of this, calling both quantities "entropy" creates a mismatch. Indeed, based on the curvature, it is natural to call the *negative* of (3) the *relative entropy*. The mismatch is also in Python: `entropy(p)` returns the concave H, but `entropy(p,q)` returns the convex I.

This confusion can be dissolved by remembering entropy and information are opposites: entropy is concave and information is convex. More generally, this error is a common psychological conflation: When measuring

[3] A proper lower semicontinuous function has a minimizer.

the temperature, are we measuring how cold it is, or how hot it is? Of course we are doing both, but this doesn't mean cold equals hot. In fact, cold equals minus hot.

While many authors' terminology conflicts with ours, some authors align partially with our usage. Schervish, in his 1995 text [19], uses "Kullback-Leibler information" for I, which is more descriptive than the widely used but opaque "Kullback-Leibler divergence" [1]. Kullback himself, in his 1959 monograph [14], refers to I as an "information function". Donsker-Varadhan, in their 1975 large deviations paper [8], denote their rate function by I rather than H; the relative information is a special case of their I.

Regarding Doob's 1934 proof, this was at the dawn of abstract probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of Kolmogorov's 1933 foundational extension theorem [13].

5. Multivariate Normal

Let μ be in $\mathbf{R}^d$, Q a positive definite symmetric $d\times d$ matrix, and $\theta=(\mu,Q)$. Let $v\cdot w$ be the dot product in $\mathbf{R}^d$. With x in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(x,\theta)=e^{-S(x,\theta)}$ where

$$S(x,\theta)=\frac{1}{2}\log\det(2\pi Q)+\frac{1}{2}(x-\mu)\cdot Q^{-1}(x-\mu). \tag{8}$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter $\theta^*=(\mu^*,Q^*)$* if its moment-generating function is

$$E_*\left(e^{v\cdot X}\right)=\exp\left(v\cdot\mu^*+\frac{1}{2}v\cdot Q^*v\right). \tag{9}$$

Suppose X is normally distributed with parameter $\theta^*=(\mu^*,Q^*)$ as in (9), and let X_1, X_2, ... be independent and identically distributed copies[4] of X. We verify **A**, **B**, **C** for this setting.

Assumptions **A** and **C** both follow from (8). To establish **B**, we need to work through some matrix manipulations [12, Chapter 2].

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v\otimes w$ is the matrix whose ij-th entry is v_iw_j. Then $v\otimes v$ is symmetric,

$$v\cdot Qv=\operatorname{trace}(Q(v\otimes v)),\qquad \operatorname{trace}(AB)=\operatorname{trace}(BA),\qquad v\otimes w+w\otimes v\le\epsilon v\otimes v+\frac{1}{\epsilon}w\otimes w, \tag{10}$$

and, by the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v\otimes v$.

Given symmetric matrices Q, R, S, we say $Q\ge 0$ if $v\cdot Qv\ge 0$ for all v, and $Q\ge R$ if $Q-R\ge 0$. If $Q\ge\delta I$ for some $\delta>0$, we say $Q>0$. If $Q\ge 0$ and $R\ge S$, then $\operatorname{trace}(QR)\ge\operatorname{trace}(QS)$.

The determinant $\det(Q)$ is the product of the eigenvalues of Q. With Q invertible symmetric and t a scalar,

$$\det(Q+tv\otimes v)=\det(Q)\left(1+tv\cdot Q^{-1}v\right). \tag{11}$$

If further $Q+tv\otimes v$ is invertible, then

$$(Q+tv\otimes v)^{-1}=Q^{-1}-\frac{t}{1+tv\cdot Q^{-1}v}\cdot(Q^{-1}v)\otimes(Q^{-1}v).$$

Lemma 1. *If $S>0$ and $Z>0$, then*[5] $\det(SZS)=\det(Z)\det(S)^2$.

Proof. Assume the result holds for Z. By (11),

$$\begin{aligned}\det(S(Z+v\otimes v)S)&=\det(SZS+Sv\otimes Sv)\\&=\det(SZS)(1+Sv\cdot(SZS)^{-1}Sv)\\&=\det(Z)\det(S)^2(1+v\cdot Z^{-1}v)=\det(Z+v\otimes v)\det(S)^2,\end{aligned}$$

hence the result holds for $Z+v\otimes v$. If $Z=\delta I$, the result holds. Since every $Z>0$ is a sum of δI and a sum of matrices of the form $v\otimes v$, the result follows by induction over the number of terms in the sum. □

[4] This setup is a special case of the Kolmogorov extension theorem.

[5] With $\det(Q)$ the product of the eigenvalues, this is not completely standard.

Define $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$ and

$$F(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \mathrm{trace}(Q^{-1} f(\mu)). \tag{12}$$

In (9), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$E_*(X) = \mu^*, \qquad E_* \left((X - \mu) \otimes (X - \mu)\right) = f(\mu). \tag{13}$$

By (13), $S(X, \theta)$ is integrable, $E_*(S(X, \theta)) = F(\mu, Q)$, and

$$I(\theta^*, \theta) = E_*(S(X, \theta) - S(X, \theta^*)) = F(\mu, Q) - F(\mu^*, Q^*).$$

Lemma 2. *Let $h(\lambda) = \log \lambda + 1/\lambda$ for $\lambda > 0$. Then $h(\lambda) \geq 1$ and $h(\lambda) = 1$ only if $\lambda = 1$. Moreover with $\epsilon = e^{-c}$, $h(\lambda) \leq c$ implies $\epsilon \leq \lambda \leq 1/\epsilon$.*

Proof. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq 1/\epsilon$. Since $\lambda \log \lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1 - 1/e)/c \geq e^{-c} = \epsilon$. ☐

Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1} Q S^{-1}$ and let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Z. Then $\mathrm{trace}(Q^{-1} Q^*) = \mathrm{trace}(S Q^{-1} S) = \mathrm{trace}(Z^{-1})$. Since $f(\mu) \geq Q^*$, by Lemmas 1 and 2,

$$\begin{aligned} 2F(\mu, Q) - 2F(\mu^*, Q^*) &\geq \log \det(SZS) - \log \det(S)^2 + \mathrm{trace}(Z^{-1} - I) \\ &= \log \det Z + \mathrm{trace}(Z^{-1} - I) = \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0. \end{aligned}$$

Since this chain of inequalities is an equality only when $Q = Q^*$ and $\mu = \mu^*$, θ^* is the true parameter.

Since $\mathrm{trace}(f(\mu)) = \mathrm{trace}(Q^*) + |\mu^* - \mu|^2$, $\mathrm{trace}(f(\mu))$ is proper. Moreover, $f(\mu) \geq \delta I$ for some $\delta > 0$.

Lemma 3. *Let $f(\mu)$ be a symmetric matrix-valued function such that trace$(f(\mu))$ is proper and $f(\mu) \geq \delta I$ for some $\delta > 0$. Then $F(\mu, Q)$ in (12) is proper.*

Proof. Since $F(\mu, Q)$ is proper iff $F(\mu, \delta Q)$ is proper, and, up to a constant, $F(\mu, \delta Q)$ corresponds to $f(\mu)/\delta$, we may assume $\delta = 1$. Now suppose $2F(\mu, Q) \leq c$. It is enough to establish $\epsilon I \leq Q \leq I/\epsilon$ for some ϵ, since this confines Q to a compact subset of the space of positive definite matrices, and implies

$$c \geq 2F(\mu, Q) \geq \log \det(2\pi \epsilon I) + \epsilon \, \mathrm{trace}(f(\mu)),$$

which confines μ to a compact subset of $\mathbf{R}^d$.

Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q. Since $f(\mu) \geq I$,

$$\sum_{i=1}^d h(\lambda_i) = \log \det(Q) + \mathrm{trace}(Q^{-1}) \leq 2F(\mu, Q) - \log \det(2\pi I) \leq c. \tag{14}$$

Since $h(\lambda) \geq 1$, for each i, $h(\lambda_i) \leq c$, hence $\epsilon \leq \lambda_i \leq 1/\epsilon$, establishing $\epsilon I \leq Q \leq I/\epsilon$. ☐

By Lemma 3, $F(\mu, Q)$ in (12) is proper, hence $E_*(S(X, \theta))$ is proper, hence $I(\theta^*, \theta)$ is proper.

Let $0 < \epsilon < 1/2$ and let $f_\epsilon(\mu) = (1 - \epsilon) f(\mu) - \epsilon Q^*$. By Lemma 3 again,

$$F_\epsilon(\theta) = F_\epsilon(\mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \mathrm{trace}(Q^{-1} f_\epsilon(\mu))$$

is proper.

Let $\bar{X}_n = (X_1 + \cdots + X_n)/n$ be the sample mean. By the law of large numbers and (13) with $\mu = \mu^*$, there is $N \geq 1$, depending on the outcome, such that, with probability one,

$$\left(\mu^* - \bar{X}_n\right) \otimes \left(\mu^* - \bar{X}_n\right) \leq \epsilon^2 Q^*, \qquad \text{for } n \geq N,$$

and

$$\frac{1}{n} \sum_{k=1}^n (X_k - \mu^*) \otimes (X_k - \mu^*) \geq (1 - \epsilon) Q^*, \qquad \text{for } n \geq N.$$

With $w = \bar{X}_n - \mu^*$ and $v = \mu - \mu^*$, using (10),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\bar{X}_n - \mu^*) \otimes (\bar{X}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$\frac{1}{n}\sum_{k=1}^{n}(X_k - \mu) \otimes (X_k - \mu) - \frac{1}{n}\sum_{k=1}^{n}(X_k - \mu^*) \otimes (X_k - \mu^*)$$

equals $v \otimes v - (v \otimes w + w \otimes v)$. Combined with the last two inequalities, this leads to

$$\frac{1}{n}\sum_{k=1}^{n}(X_k - \mu) \otimes (X_k - \mu) \geq f_\epsilon(\mu)$$

for all μ and $n \geq N$. Since this implies

$$F_\epsilon(\theta) \leq \bar{S}_n(\theta), \qquad \text{for all } \theta \text{ and } n \geq N,$$

this establishes **B**.

By differentiating $\bar{S}_n(\mu + t\nu, Q)$ and $\bar{S}_n(\mu, Q + tv \otimes v)$ at $t = 0$, the MLE is

$$\hat{\mu}_n = \frac{1}{n}\sum_{k=1}^{n} X_k, \qquad \hat{Q}_n = \frac{1}{n}\sum_{k=1}^{n}(X_k - \hat{\mu}_n) \otimes (X_k - \hat{\mu}_n).$$

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